Mind the Mic

Tsz Chai Alan Liu
Sonoform Labs
Ontario, Canada
tcaliu@sonoform.org

Abstract

Neural acoustic field methods aim to predict what a microphone would record at an arbitrary point inside a room and are typically evaluated using log-spectral distance at microphone positions withheld from training. In this work, we investigate how much this distance varies with receiver position through two complementary experiments. The first uses fifteen microphones mounted on a 1.82 m bar to record each hammer strike simultaneously, allowing us to compare recordings at different receiver positions while keeping the sound fixed. We find that changing only the receiver position produces a distance change of 0.1376 natural-log units under peak normalization, approximately 3.8 times the change caused by a specified preprocessing control (0.0362). Furthermore, when the 10^{-3} constant used by NAF and NeRAF is included under peak normalization, changing the point of impact on the object produces a comparable change in the metric to moving the microphone. The second experiment examines the publicly available DiffRIR benchmark across four rooms, where we first reproduce the nearest-neighbor and model results to within 0.7% in all rooms before examining how receiver separation affects the evaluation metric under the benchmark's data splits. At each room's held-out displacement (the distance between a test microphone and its nearest training microphone), the difference between two measured recordings is 7.9–10.1 times the improvement reported by DiffRIR. Since the nearest-neighbor baseline predicts using a recording from another receiver, its error reflects this same difference, with the baseline error reported in the paper alone being 7.8–10.0 times the reported model improvement. However, when both methods are evaluated at each position against the same recording, DiffRIR outperforms the baseline at 98–100% of positions in the three reverberant rooms, while underperforming it at 26.8% of positions in the dampened room. These results demonstrate that receiver position can have a greater impact on the absolute evaluation score than the reported model improvement, while the size of that improvement also varies substantially across positions.

1 Introduction

A neural acoustic field interpolates the recording a microphone would make at an arbitrary position in a room from a sparse set of measured positions within it [Luo et al., 2022, Su et al., 2022, Liang et al., 2023]. Since the goal is prediction at held-out positions, such a model is evaluated by withholding some of the measured positions from training and scoring the prediction at each against the measurement with a log-spectral distance. By that distance, on the benchmark we examine, DiffRIR [Wang et al., 2024] reduces a nearest-neighbor baseline's error from 5.99 to 5.22 in the metric's units, a reported improvement of 13 percent. In other fields, margins of that relative size have proven unstable; for instance, in language model evaluation, merely reformatting a prompt without changing its meaning produces accuracy differences of up to 76 points on a single model [Sclar

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et al., 2024], and reordering a question's answer choices changes where a model ranks by as many as eight places [Alzahrani et al., 2024]. Each result assumes the perturbation preserved the question. That assumption is easy to violate: a change that made the task harder would move the score the same way, for legitimate reasons. A key open question for evaluation in this field is therefore how much of a reported margin reflects the model and how much reflects the apparatus—the microphone. We find that the question is rarely put to the test at all. Reading each benchmark's released split files shows why: the microphones a model is scored on sit a median of 70 microns from the ones it trained on for RAF, and on SoundSpaces they are the same positions, so both splits score held-out items at receiver positions present in training.

A microphone array removes the assumption that the perturbation preserved the content. Ours is a bar of fifteen microphones spanning 1.82 m, so the measurement applies only where a benchmark withholds microphones at separations inside that span. Only DiffRIR does, at a median 0.96 m; elsewhere the median is below 0.4 m and usually zero (Section 3.3). We measure the effect of receiver position twice, first in the laboratory beside a preprocessing control that holds the microphone fixed, then in DiffRIR's metric and split beside our reproduction of DiffRIR's published results.

Contributions.

1. Moving the receiver alone moves the metric by 0.1376, about 3.8 times a specified preprocessing control (0.0362), under a design where fifteen microphones capture each hammer strike at a single instant so the source is identical across all fifteen. Ground-truth recordings are the microphone measurements the models aim to match. On DiffRIR's four rooms, two such recordings separated by that benchmark's held-out displacement differ by 7.9 to 10.1 times the improvement it reports, close to the 7.8 to 10.0 that the published nearest-neighbor error implies. Scored position by position, the model beats the baseline at 98 to 100 percent of positions in three rooms and at 73.2 percent in the dampened one.
2. Two of the four benchmarks we read score held-out items at receiver positions already in training: the released split files place held-out microphones zero meters from a training microphone on SoundSpaces and a median 0.00007 m away on RAF, while DiffRIR withholds the receiver positions themselves.
3. Half the spectrogram bins in DiffRIR's released classroom data sit at the soft floor of its metric, which adds 10^{-6} to the squared magnitude inside a square root to keep the logarithm finite, a regularization scale of 10^{-3} in magnitude. Across four papers, NAF and NeRAF share one definition of log-spectral distance, DiffRIR uses another and RAF releases no evaluation code; NeRAF labels natural-log values in decibels, understating the error by a factor of 8.7, and RAF labels its table in decibels with no released evaluator behind the label.
4. Pooling across the forty gantry configurations inflates the estimate by a factor of 1.32 to 1.45, because the object is struck again each time the gantry moves, mixing a change of microphone with a change of strike.

2 Method

2.1 Data

RealImpact [Clarke et al., 2023] records rigid objects struck by an instrumented impact hammer, one that measures the force it applies, and captured by a microphone gantry, a motorized rig that repositions the microphones between measurements. We use the released preprocessed set, which contains 3000 recordings per object: 5 strike positions on the object surface, 40 gantry configurations (10 angles around the object by 4 distances from it, 0.23 m to 1.23 m), and 15 microphones spaced 0.13 m apart on a vertical bar spanning 1.82 m.

We use 20 objects spanning five material classes (ceramic, wood, metal, plastic, glass). Recordings are the released `deconvolved_0db` signals, already divided by the measured hammer force, which removes much of the strike-force variation (Section 5).

2.2 Design

We isolate how much the metric moves when only receiver position changes. The gantry's 40 configurations appear to offer exactly that, but the object is struck again after each move, and only the 15-microphone bar captures one strike at once. The released preprocessing code ties each strike to the fifteen microphones of one bar, replicating each hammer trace and vertex identifier exactly fifteen times, once per microphone (`np.repeat(..., NUM_MICROPHONES, axis=0)`). Every comparison we make is therefore among the 15 microphones of a single bar at a single gantry configuration, where the strike, object, room and instant are identical, with only the receiver varying: its height and the capsule mounted there. Pooling across gantry positions raises the estimate by a factor of 1.32 to 1.45, depending on the metric.

2.3 Statistic

The metric is used to compare two candidates against a reference; we evaluate one such comparison at each of the fifteen microphones and measure how much its outcome varies across the microphones. For one object at one gantry configuration, let r be the reference strike position and i and j two others. For each microphone m in the bar, compute

$$\delta(m) = d\big(x_r(m),\, x_i(m)\big) - d\big(x_r(m),\, x_j(m)\big), \tag{1}$$

where d is the metric under test and $x_s(m)$ is the recording of strike s at microphone m. Both distances at a given m use recordings made through the same microphone, so an ideal multiplicative microphone response cancels inside each distance, up to the regularizing constant and the per-recording scaling; capsule identity and position stay confounded (Section 5). The residual is how much the gap between two sources depends on the receiver. We report

$$\sigma = \mathrm{sd}_m\big[\delta(m)\big], \tag{2}$$

the standard deviation of that paired difference over the 15 microphones. We compute σ for every object, at 5 of the 10 azimuths and all 4 distances, and for every reference strike with every pair of the other four, 30 comparisons per cell. That gives 12,000 values of σ, and we report the median with a 95 percent interval from 4000 bootstrap resamples drawn over whole objects, since the values are nested within them.

We leave σ unstandardized so it compares directly against numbers in published results tables.

2.4 Metrics

All three metrics run at 48 kHz on 0.5 s excerpts, log-spectral distance (LSD), log-mel and MFCC, each defined in Appendix A. Each excerpt's onset is located by tracking energy over a 64-sample window, and the excerpt starts 200 samples before the onset, scaled so its loudest sample is 1. LSD values are natural-log magnitude differences, a factor of 8.686 from decibels; log-mel is a natural log of power, a factor of 4.343, and MFCC inherits librosa's decibel scaling. We convert only when a paper states its metric completely, and mark the comparison unverified when the paper does not.

3 Results

3.1 Isolating Receiver Position

Moving the microphone alone moves the metric by 0.1376 natural-log units, about 3.8 times the 0.0362 of a specified preprocessing control that holds the microphone fixed and changes only the processing. The value 0.1376 is a standard deviation of the paired log-spectral difference over $n = 12{,}000$ comparisons, 20 objects and 400 object-configuration cells, 5 of the 10 azimuths by all 4 distances per object (95 percent CI $[0.1158, 0.1634]$, resampling objects). The control re-processes one microphone's recording fifteen times, varying the 200-sample lead-in by ± 100 samples and alternating peak with RMS scaling, on the twelve-object subset described below, where the receiver figure is 0.151 and the ratio 4.2; the control covers these two choices alone. All three metrics appear in Table 4 and Figure 3, Appendix D, where the control reaches 23 to 24 percent of the receiver figure on the same twelve objects.

Alignment and window length move the twelve-object receiver figure of 0.151 by at most 8 percent: sound reaches the fifteen microphones at different times, so cross-correlation alignment before comparison removes that timing difference and still returns 0.157, ruling out propagation delay across a bar whose worst-case path difference is 0.940 m or 2.74 ms, while doubling the analysis window to 1.0 s gives 0.163. Normalization moves it more: switching from peak to RMS scaling lowers the figure by a third, from 0.151 to 0.102, still 2.8 times the control. The headline uses peak normalization, the larger of the two, so every magnitude in this section belongs to that pipeline.

Across object classes the effect spans a factor of 2.2, from wood at 0.218 down to ceramic at 0.097, with plastic at 0.173 and metal at 0.151 between them and glass unrepresented in this subset. The size of σ therefore depends on the material.

3.2 Signal Against Nuisance

Moving the microphone changes the metric about as much as striking a different position on the same object, a signal-to-nuisance ratio of 0.97. Signal and nuisance both start from the distances $d(x_r(m), x_i(m))$ between a reference strike r and each of the four other strikes i. The signal is their standard deviation over the four strikes after averaging over microphones (0.01660); the nuisance is their standard deviation over the fifteen microphones, averaged over strikes (0.01713). The σ of Section 2 is the spread over microphones of a difference of two such distances, and the nuisance is the spread of one. Both are medians over 256 cells (16 objects, 4 azimuths, 4 distances) of a mean absolute log-magnitude difference under scipy's STFT scaling, with 10^{-3} added to each magnitude of the peak-normalized excerpts, the constant NAF and NeRAF use (Figure 4, Appendix D).

Two things keep that 0.97 from showing whether the nuisance exceeds the signal. The constant moves the ratio: the ratio is 1.15 when the constant is 10^{-6}, and the larger constant shrinks the signal by $7.84\times$ and the nuisance by $6.58\times$, so the ratio falls 16 percent. Sampling is uneven: the median of a sample standard deviation is biased low by 11 percent with three of four strike positions free to vary, against 2.4 percent with fourteen of fifteen microphones, and correcting that asymmetry returns the ratio above one. Either way signal and nuisance are of the same magnitude.

3.3 Displacement Across Benchmarks

In three of the four benchmarks, the median held-out receiver sits within 0.4 m of a training receiver, and in two of them within a centimeter. Under each official split, the median distance to the nearest training receiver is 0.000 m on SoundSpaces [Chen et al., 2020], whose test receivers appear in training [Luo et al., 2022, Su et al., 2022, Liang et al., 2023]; 0.00007 m for RAF's 80/5/15 split [Chen et al., 2024] (worst case 0.0095 m, $n = 7{,}123$); roughly 0.1 to 0.4 m for RWAVS [Liang et al., 2023], whose unpublished scene scale is estimated; and 0.959 m for DiffRIR's `classroomBase` (range 0.255 to 1.891 m, $n = 463$). These distances concern receivers alone: SoundSpaces work holds out new emitter-listener combinations [Luo et al., 2022] and RAF moves the speaker while its microphone tower stays [Chen et al., 2024], whereas DiffRIR withholds receiver positions. Only DiffRIR approaches the bar's range, so it alone receives a number; Appendix B gives sources and computation.

3.4 Measuring on DiffRIR

Moving the receiver alone changes the metric by 7.9 to 10.1 times the improvement DiffRIR reports, in all four rooms. The measurement uses their multiscale log-L1 metric [Wang et al., 2024], computed in their implementation, units and evaluation length. The measurement runs on the released ground-truth room impulse responses, each a room's answer to a single instantaneous sound and the measured recording a model is scored against, and on renderings of their weights at every held-out position. Their baseline is one such recording captured a median 0.74 to 1.26 m away, so its error is itself a distance between receivers, and the reported improvement retires 10 to 13 percent of that error; published nearest-neighbor error over gain alone gives 7.8, 9.1, 10.0 and 8.4. The measurement adds the size of that distance at the held-out separations, a property of the sound field the model predicts; Section 3.5 carries the evidence about the model.

The measurement depends on first reproducing the published results. The strongest baseline is nearest-neighbor prediction, which predicts an unseen microphone's recording by copying the

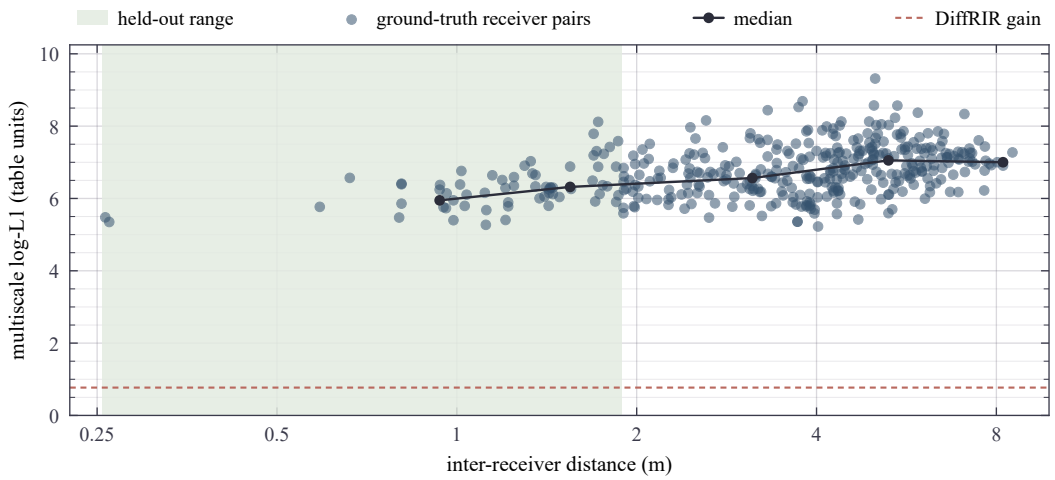


Figure 1: Each dot is a pair of ground-truth recordings from DiffRIR's classroom data. The horizontal axis gives the distance between the two microphones, the vertical the multiscale log-L1 between their recordings, and the solid line shows the median of pairs grouped by distance. The shaded band marks the held-out range, 0.255 to 1.891 m, where the benchmark withholds receivers from training and scores models on them; the closest pair sampled is 0.26 m apart. The dashed line at 0.77 is the improvement DiffRIR reports for this room. That improvement is a difference between two systems at one position, while each dot is a distance between two positions, so that dashed line is a scale reference; Section 3.5 makes the matched comparison. Pairs are drawn uniformly at random with a fixed seed, 399 here and 150 per room within the band for Table 1.

recording of the closest training microphone. Under DiffRIR's split code, our reimplementation of that baseline and a rendering of the released weights match the published table in every room of the four, all eight numbers within 0.7 percent and seven of eight within 0.4 percent.

Table 1: Reproduction and between-receiver distance on all four DiffRIR rooms. Columns 2–5 validate the pipeline by setting our reimplementation of each method against the number DiffRIR published. Gain is the published improvement, the nearest-neighbor value minus the model value, and band gives the range of separations between test and training receivers in each room. Between receivers is the median metric value over 150 pairs of ground-truth receiver positions sampled at those distances, and the last column reports it as a multiple of the gain.

	nearest neighbor		DiffRIR			between receivers		
room	reproduced	published	reproduced	published	gain	band (m)	median	÷ gain
classroom	5.99	5.99	5.20	5.22	0.77	0.25–1.89	6.10	7.9×
dampened	1.36	1.36	1.22	1.21	0.15	0.11–1.43	1.51	10.1×
hallway	10.14	10.14	9.12	9.13	1.01	0.30–1.71	10.18	10.1×
complex	5.52	5.52	4.85	4.86	0.66	0.39–2.63	5.68	8.6×

The ratio is taken inside the band each room holds out, and dropping that restriction raises every ratio, to between 8.6 and 12.8 times. All 150 sampled pairs in each room exceed the reported gain, a statement about that sample. The unrestricted classroom median holds at 6.62 to 6.70 across three sampling seeds. Per-room values are in Table 1, and Figure 1 plots the classroom distribution against separation.

3.5 Per-Position Comparison

The DiffRIR model beats the nearest-neighbor baseline at every held-out position in the hallway and nearly every position in the other reverberant rooms, while in the dampened room the baseline beats it at 26.8 percent of positions. For each held-out position, the model's error and the baseline's error are computed at that same position, and the difference between them is the per-position gain.

Table 2: Paired per-position comparison of the DiffRIR model against the nearest-neighbor baseline, under the multiscale log-L1 metric. Both are scored at each of the n held-out positions in a room, and the last column counts the positions where the baseline has the lower error, with the share and a 95 percent Clopper-Pearson interval [Clopper and Pearson, 1934] that treats the n tested positions as independent; neighboring positions are correlated, so the interval describes these positions only.

room	n	per-position gain, mean (sd)	nearest-neighbor beats DiffRIR
hallway	423	1.02 (0.26)	0 (0.0%) [0.00, 0.87]
classroom	463	0.79 (0.53)	6 (1.3%) [0.48, 2.80]
complex	198	0.67 (0.40)	4 (2.0%) [0.55, 5.09]
dampened	198	0.15 (0.21)	**53 (26.8%) [20.74, 33.51]**

Measured this way the ranking is resolved in every room: treating positions as independent, the mean per-position gain over its standard error gives paired t values from 9.6 in the dampened room to 81.6 in the hallway. The margins are heterogeneous. Their standard deviation exceeds the mean in the dampened room, where the model wins at 145 of 198 positions and the baseline at the other 53 (Table 2).

Under DiffRIR's envelope distance, a measure of how fast the sound decays, the baseline beats the model at 21.6 to 40.4 percent of positions in every room, though the mean improvement is evident in all four rooms. The hallway is the clearest case: the model wins at every position under the spectral metric and loses at 40.4 percent of them under the envelope. Two ground-truth recordings at different receivers differ under the envelope distance by a median of 7.5 to 38 times the reported envelope gain over unrestricted pairs, and 6.7 to 34 inside the held-out band, close to the ratio of the nearest-neighbor envelope error to that gain. All of these envelope statistics are derived from a reproduction which matches the nearest-neighbor column to within 0.5 percent and the model row to within 2.1 percent. In the hallway the model mismatch is 1.4 percent, 84 percent of the reproduced envelope gain, which is about half the published one.

3.6 Metric Floor

DiffRIR's metric places half the short-time Fourier transform bins in the classroom data at a soft floor of 10^{-3} in magnitude. Those bins sit within 0.08 percent of that floor at every window size, and 92.5 to 99.1 percent of all bins fall within a factor of ten of it (Figure 2, Appendix D). The floor traces to the implementation: DiffRIR adds $\varepsilon = 10^{-6}$ to the squared magnitude inside the square root, which sets a regularization scale of $\sqrt{10^{-6}} = 10^{-3}$ in magnitude. The mapping is smooth, so differences below that scale are compressed and still counted.

An absolute floor depends on recording gain. DiffRIR does not normalize recordings, so the same metric takes different values on recordings with different signal scales. The name log-spectral distance is ambiguous as well. NAF and NeRAF compute the same mean absolute difference of $\ln(|X| + 10^{-3})$, DiffRIR computes the multiscale form above, and RAF releases no evaluation code [Luo et al., 2022, Brunetto et al., 2025, Wang et al., 2024, Chen et al., 2024]. NeRAF labels natural-log values in dB, which understates the error by a factor of 8.7: the label heads its Table 1 while the evaluator skips the conversion. RAF also labels its table in dB, and with no released evaluator its base stays unverified.

3.7 Reporting Practice

The variation measured above would be invisible in the fifteen papers we surveyed. All but one report no confidence interval, standard error, standard deviation or significance test on the objective metric carrying their main claim, and three decline the error-bar item on their venue checklist. Appendix C lists the survey and names the exception.

4 Related Work

Measurement resolution. The quantity measured above has a long precedent in instrument metrology, which Welty et al. [2019] import into machine learning evaluation, observing that papers

routinely compare two systems without reporting whether the performance difference lies within the instrument's capacity to measure it. That observation takes formal shape in generalizability theory [Cronbach et al., 1972], which splits a measurement's variance into the facets that produced it, and in discriminant validity [Campbell and Fiske, 1959], which asks whether an instrument responds more to its target than to an irrelevant factor. Several subfields have already acted on that observation, reporting score distributions, decomposing benchmark variance, measuring statistical power and re-examining conclusions drawn from a handful of runs [Reimers and Gurevych, 2017, Bouthillier et al., 2021, Card et al., 2020, Agarwal et al., 2021].

Audio thresholds. The closest prior work in our field is Urbano et al. [2012], which reports that a difference of at least 0.3 is needed before users prefer the supposedly better system more than half the time. The same author also applies generalizability theory to MIREX, the annual music information retrieval evaluation campaign [Urbano, 2013], and Urbano et al. [2014] run a random-effects variance decomposition on *distances from a fixed reference recording to altered versions*, with codec, bitrate and frame size as channel facets. That is structurally our design, with lossy encoding where we have microphone geometry. Outside music retrieval, room acoustics standardized thresholds long ago: ISO 3382-1:2009 [International Organization for Standardization, 2009] tabulates just-noticeable differences, the smallest change a listener can hear, and Lubman [1969] derives in closed form how many microphone positions a measurement needs when sound arrives from all directions roughly equally.

Prompt sensitivity. The figures quoted in Section 1 come from this literature [Sclar et al., 2024, Alzahrani et al., 2024], and Mizrahi et al. [2024] measure how far single-prompt results shift across 20 models and 39 tasks, and call for multi-prompt evaluation. Each of them relies on the underlying content surviving the perturbation, which remains an assumption, and Hua et al. [2025] argue that a substantial part of measured prompt sensitivity originates in scoring. In our design fifteen microphones record one sound, so any difference among the fifteen recordings comes from the receiver alone.

Ranking stability. Voorhees [2000] varies one factor and measures the effect on scores, as we do: relevance judgments differ substantially between assessors while system rankings stay stable. Voorhees and Buckley [2002] derive empirical error rates for how often a different set of topics of the same size would reverse a conclusion. An assessor change applies roughly uniformly across systems, so it cancels in a ranking, as a shared microphone cancels inside our paired difference.

Audio metrics. Parmar et al. [2022] show in vision that resizing and compression can induce large variations in FID while leaving an image perceptually unchanged. Within this community, Acoustic Volume Rendering [Lan et al., 2024] argues that spectral-magnitude metrics capture only part of the waveform information; log-spectral distance belongs to that class. No published work establishes a threshold for per-item log-spectral distance at held-out receiver positions.

5 Limitations

Separation range. Our receivers span 0.13 to $1.82\,\mathrm{m}$, and Section 3.3 shows that SoundSpaces and RAF hold out receivers orders of magnitude closer. At those separations our σ overstates the receiver effect, so we restrict our numerical claims to DiffRIR, the one benchmark with a receiver split as wide as ours. Figure 1 reports the metric between ground-truth pairs as a function of separation from $0.26\,\mathrm{m}$ upward in the classroom, a range that covers nearly all of that room's held-out band and excludes the sub-centimeter splits.

Source definition. A source here is one of five strike positions on one object. We measure how the metric responds to a change of source with the receiver held fixed, as a proxy for the model-to-model difference a benchmark would report. Those five positions may span a smaller or larger difference than a typical model-versus-model gap, so the proxy holds only if moving the microphone perturbs any comparison by a similar amount, whatever two signals are compared. The five positions also yield only ten unordered pairs per object, the sample the signal side of the ratio depends on.

Scope. The controlled measurements come from one acoustically treated space and one hammer, and our numbers may not carry to rooms and sources that differ. Each microphone height has a different capsule, so capsule identity and position remain confounded; the paired difference cancels only an ideal multiplicative capsule response. Sections 3.4 and 3.5 partially address the one-space, one-hammer scope with a second dataset, four more rooms and a second metric implementation. The strike itself also varies: deconvolution removes much of the strike-force variation but leaves contact point, contact angle and hammer rebound. The median σ of 0.1376 avoids these by comparing only within a single gantry configuration, where all fifteen microphones record one strike. Any quantity pooled across gantry configurations picks up the strike variation, since the object is struck again each time the gantry moves.

Open gaps. Our σ states how much the metric moves when the receiver moves; whether a listener would hear a difference that size is untested. Separately, RAF's table is labeled dB, RAF releases no evaluation code, and its constant is unpublished, so any comparison of σ with RAF's table remains unverified.

Falsification. Striking more positions per object would test the signal-to-nuisance result, and scoring several trained models at the same positions would test whether Table 2 holds beyond one model pair.

6 Conclusion

Neural acoustic field models are ranked by a distance computed at microphone positions withheld from training, and progress in the field is measured against that number. How much of that number tracks the model and how much tracks the receiver positions decides how a reported gain should be read.

We measured how much the log-spectral distance between two recordings moves when the microphone moves. Across fifteen microphones recording each strike at once, that movement is 0.1376 in natural-log units, about 3.8 times a specified preprocessing control. On the one benchmark whose held-out microphones sit as far from training as our range, the distance between receivers is 7.9 to 10.1 times the reported improvement in all four rooms, close to the 7.8 to 10.0 that the nearest-neighbor error already implies. Receiver position therefore changes the absolute score by more than the model gain. Position by position, the model keeps its ranking at 98 to 100 percent of positions in three rooms, and the baseline wins at 26.8 percent of positions in the dampened room.

Authors reporting this metric should therefore print the per-position standard deviation of the paired gain beside its mean, and state the exact definition of log-spectral distance they used, with its log base and constant.

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A Metric definitions

All metrics are computed at 48 kHz on 0.5 s excerpts, onset-located by a 64-sample energy envelope with 200 samples of pre-roll, peak normalized unless stated otherwise.

Table 3: Metric definitions. LSD is a natural-log magnitude difference, which converts to decibels by a factor of 8.686; log-mel is a natural log of power, a factor of 4.343; MFCC inherits librosa's decibel scaling.

metric	definition as computed here
log-spectral distance (LSD)	root-mean-square difference of the natural log of short-time Fourier transform magnitudes plus 10^{-6}, that is the energy in each frequency band in each short time window; scipy scaling, 512-sample window, 75 percent overlap
log-mel	as above but on a mel spectrogram, which groups frequencies into bands spaced to match human pitch perception; 64 bands, 512-sample window, 128-sample hop
MFCC	as above on mel-frequency cepstral coefficients, a compact 20-number summary of spectral shape widely used in speech systems

Sections 3.4 and 3.5 compute DiffRIR's multiscale log-L1 metric with the released DiffRIR code: the mean absolute difference of $\log \sqrt{\mathrm{re}^2 + \mathrm{im}^2 + 10^{-6}}$, summed over window lengths $\{64, 128, 256, 512, 1024, 2048, 4096\}$ with hop $n/4$ and a Hann window, on the first 96,000 samples (2.0 s) of each impulse response, which is their evaluation length, and multiplied by 10, the scale of their published table. Using the full 14 s recording dilutes every result by a factor of 5.2, since the added tail sits at the floor. That dilution is the easiest way to fail to reproduce their table.

B Held-out displacement survey

For each benchmark we take the official split as released and compute, for every held-out receiver, the Euclidean distance to the nearest training receiver. SoundSpaces-based work (NAF, INRAS, AV-NeRF) evaluates at receiver positions that also appear in training, giving zero displacement; NAF describes its held-out unit as a new combination of emitter and listener, so these splits test new source-receiver combinations. RAF's official 80/5/15 split gives a median of 0.00007 m over $n = 7{,}123$ held-out receivers, worst case 0.0095 m; RAF moves the speaker while its microphone tower stays, so the receiver displacement alone leaves open whether a held-out source-receiver combination appeared in training. AV-NeRF evaluates on RWAVS, whose coordinates are in an unpublished scale; converting by the scene scale estimate gives roughly 0.1 to 0.4 m, and we mark that range approximate. DiffRIR's `classroomBase` split trains on 12 of $n = 630$ receivers (`train_indices = np.arange(12) * 57`) and leaves 618, of which 155 form a validation split and 463 the test split. The 463 test receivers sit a median 0.959 m from the nearest training receiver, range 0.255 to 1.891 m. DiffRIR is the only benchmark whose held-out displacement overlaps the 0.13 to 1.82 m span of the RealImpact microphone bar, and therefore the only one against which we place a number.

C Reporting-practice survey

We surveyed fifteen papers spanning neural acoustic fields, novel-view acoustic synthesis and neural modal sound synthesis, selected by hand, so they form a convenience sample. Ten are acoustic-field or novel-view papers: NAF [Luo et al., 2022], INRAS [Su et al., 2022], Few-Shot Audio-Visual Learning of Environment Acoustics [Majumder et al., 2022], AV-NeRF [Liang et al., 2023], Novel-View Acoustic Synthesis (CVPR 2023, arXiv 2301.08730), RAF [Chen et al., 2024], DiffRIR [Wang et al., 2024], NeRAF [Brunetto et al., 2025], AV-GS (NeurIPS 2024, arXiv 2406.08920) and Acoustic Volume Rendering [Lan et al., 2024]. Five come from impact and modal sound synthesis: NeuralSound (SIGGRAPH 2022, arXiv 2108.07425), DiffSound (SIGGRAPH 2024, arXiv 2409.13486), DiffImpact (CoRL 2021, PMLR volume 164), RealImpact [Clarke et al., 2023] and AV-MSF (arXiv 2608.05145). Fourteen report no confidence interval, standard error, standard deviation or significance test on the objective metric carrying their principal claim.

Few-Shot Audio-Visual Learning of Environment Acoustics [Majumder et al., 2022] is the exception: it asserts $p \leq 0.05$ with the test unnamed and no interval printed. INRAS, AV-GS and Acoustic Volume Rendering explicitly decline the error-bar item on their venue checklist, and INRAS and Acoustic Volume Rendering cite computational cost.

The sharpest single case is NeRAF [Brunetto et al., 2025], whose Section 4.2 states that each experiment is run six times and averaged "to ensure statistically significant results", and whose tables then print single numbers with no dispersion. The spread across those six runs goes unreported, the practice Agarwal et al. [2021] identify as having changed published conclusions elsewhere.

For completeness: RealImpact [Clarke et al., 2023] does report relative standard deviations over ten measurement trials and peak-force standard deviations of 29.8 N and 11.7 N. Those figures characterize the dataset.

D Supplementary table and figures

Table 4: Receiver-position σ against the specified preprocessing control. The control holds one microphone fixed and varies only the 200-sample lead-in and the choice of peak or RMS scaling, on a twelve-object subset; the last column divides it by the receiver figure on that subset. Intervals here resample the pooled comparisons and so describe precision within the sampled objects; resampling whole objects widens the LSD interval to $[0.1158, 0.1634]$, which is the interval quoted in Section 3 and still exceeds the control by more than threefold.

metric	σ, receiver moves	95% CI (pooled)	control	control / receiver (12 objects)
LSD	0.1376	[0.1358, 0.1395]	0.0362	0.24
log-mel (power)	0.3581	[0.3544, 0.3620]	0.0839	0.23
MFCC	4.148	[4.104, 4.195]	1.048	0.24

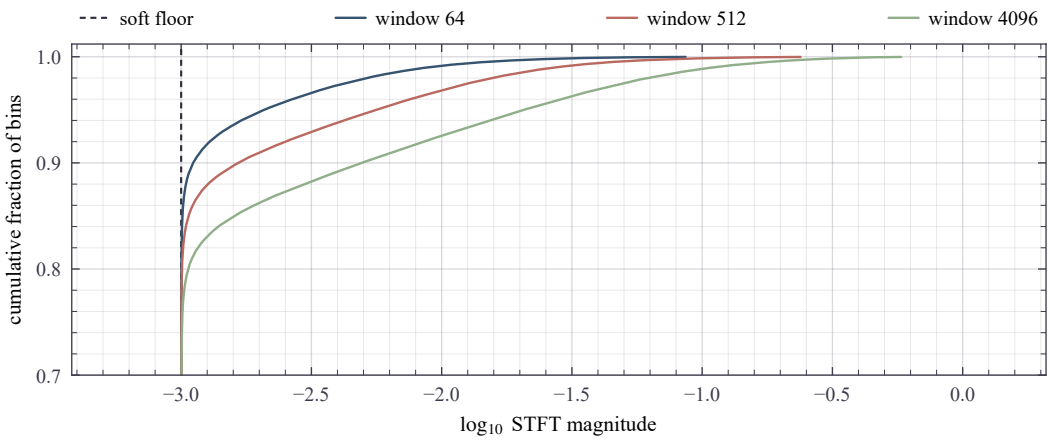


Figure 2: Cumulative distribution of STFT magnitudes on DiffRIR's released classroom data under their transform, over all 630 impulse responses, roughly 122 million bins per window; the curves show three of the seven window lengths. The median bin sits within 0.08 percent of the soft floor of $\sqrt{10^{-6}} = 10^{-3}$ at every window size from 64 to 4096 (1.00001×10^{-3} to 1.00073×10^{-3}), and between 92.5 and 99.1 percent of bins fall within a factor of ten of it, the fraction falling monotonically as the window lengthens. The vertical axis begins at 0.70; every curve rises from zero at the floor itself.

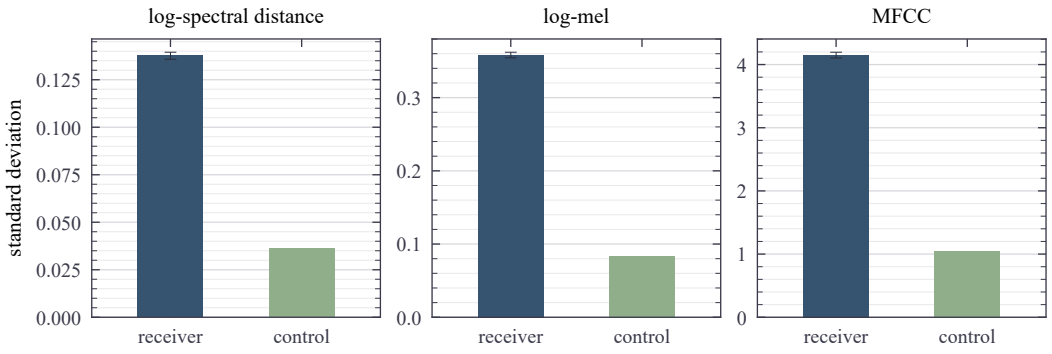


Figure 3: Receiver-position σ against the specified preprocessing control, per metric. Error bars are the pooled 95 percent intervals of Table 4; resampling objects widens the LSD interval to $[0.1158, 0.1634]$. The control holds one microphone fixed and varies only the 200-sample lead-in and the choice of peak or RMS scaling, reaching 23 to 24 percent of the receiver figure on the same twelve objects; the receiver bars show all 20 objects.

E Reproduction scripts

Both datasets are public, and the code, split indices and the reporting-practice coding sheet are available at `https://github.com/Sonoform-Labs/mind-the-mic`. The displacement survey (Section 3.3) and the reporting-practice counts (Section 3.7) were compiled by hand from the cited releases and papers. Table 2's intervals and the paired t values follow from the counts, means and standard deviations in `out_perposition.json`. All other measured numbers, figures and tables regenerate from sixteen scripts: `m_within_column.py` (microphone σ), `m_cluster_boot.py` (object-level cluster bootstrap on that σ), `m_df_bias_check.py` (degrees-of-freedom bias in the ratio estimator), `m_pathdiff.py` (worst-case within-column path difference), `m_controls.py` (alignment, window length, normalization, per-material), `m_snr.py` (signal versus nuisance under both floor conventions), `m_grid.py` (aggregation conventions and sample rates), `m_diffrir_final.py` (four-room unrestricted ratios), `m_matched_displacement.py` (in-band ratios of Table 1), `m_perposition.py` (reproduction and per-position comparison, using DiffRIR's data loader and split code), `m_env_perposition.py` (same under the envelope metric),

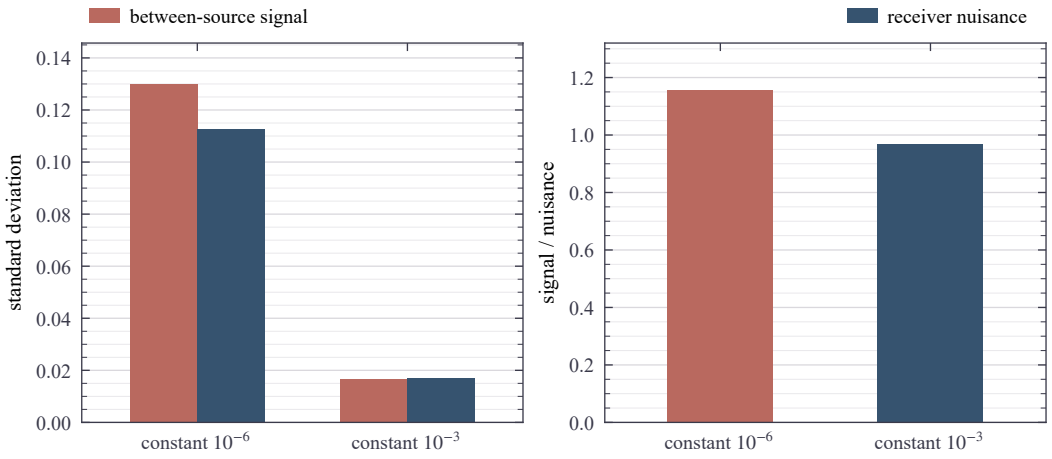


Figure 4: Left: between-source signal and receiver nuisance (Section 3) with a constant of 10^{-6} and with the 10^{-3} soft floor of NAF and NeRAF; the larger constant shrinks both by roughly an order of magnitude. Right: their ratio, against the gridline at parity. The ratio moves from 1.155 to 0.969, so with the constant NAF and NeRAF use the two quantities are of the same order. Bar color marks which quantity is larger.

`m_env_band.py` (envelope ratios over unrestricted and in-band pairs), `m_backup_controls.py` (full-test-set confidence interval, three-seed stability), `m_floor_verify.py` (floor statistics of the Metric Floor subsection and the values in the caption of Figure 2), `m_reproduce.py` (full-length classroom reproduction behind the dilution factor in Appendix A), and `m_figures.py` (all four figures).